



Introduction

Mental health apps are widely used to support emotional well-being. However, star ratings alone fail to capture the nuanced emotions expressed in user reviews.

This project applies sentiment analysis to Arabic-language reviews of mental health applications, providing insights into user perceptions in the underrepresented Arabic-speaking market.

Objectives

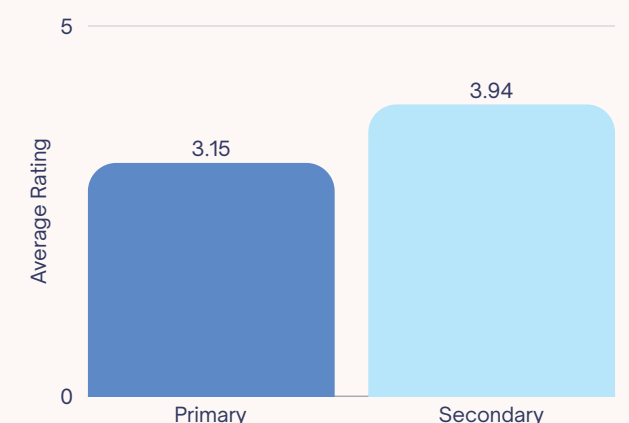
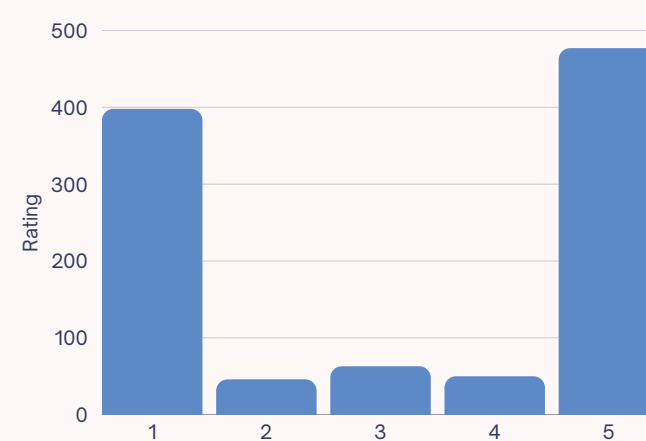
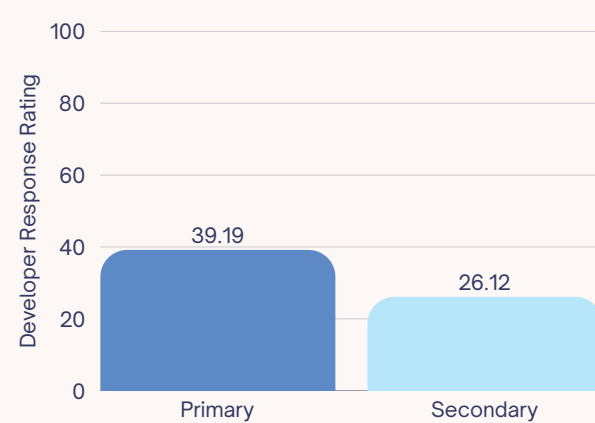
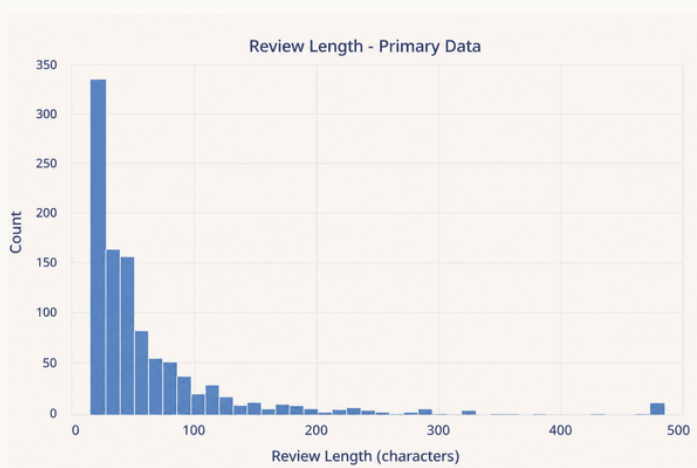
- **How can sentiment analysis help understand Arabic-speaking users' perceptions of mental health apps?**
- Can machine learning models effectively classify Arabic reviews into positive, negative, and neutral sentiment?
- What challenges affect sentiment classification performance in Arabic mental health app reviews?

Data Collection

- Scraped from Google Play Store using Python (Arabic filter).
- 4 apps (Labayh, Estanara, Calm, BetterHelp) — 1,032 reviews
- Data includes: review text, rating, date, likes, developer responses
- Data was cleaned and prepared for analysis
- Secondary dataset: MHARD (18,000 English reviews) for comparison.

Data Analysis

- Ratings show a **polarized** pattern — most reviews are 5-star or 1-star; users engage at extremes.
- Arabic users show **lower average satisfaction** (3.15) vs. global (3.94).
- Lower-rated reviews tend to be longer (correlation ≈ -0.13).
- **39%** developer response rate in **Arabic** dataset; **26%** in global.



Models and Findings

- Built 3 models: Logistic Regression, Random Forest, XGBoost
- Used TF-IDF for feature extraction
- Evaluated using Accuracy and Macro F1-score
- Best model: Logistic Regression
- (Accuracy: 78%, Macro F1: 0.54)
- Other models achieved $\sim 71\%$ accuracy
- Neutral class was hardest to classify

Logistic Regression (Baseline)				
True label	negative	neutral	positive	
	negative	76	0	13
	neutral	5	0	8
	positive	19	0	86
		negative	neutral	positive

Conclusions

- Results show polarized feedback (mostly very positive or very negative)
- Arabic users tend to show lower satisfaction compared to global users
- Lower-rated reviews tend to be longer and more detailed
- Developer response is higher in Arabic apps

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